

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:50

Annexure A

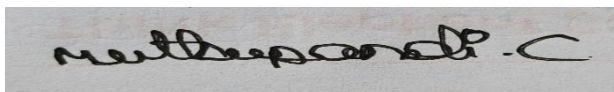
Technical Presentation

Criteria	Technical Content Accuracy(3)	Problem Identification(2.5)	Use of Tools(2)	Communication(1.5)	Professionalism(1)	Total(10)
Weightage	30%	25%	20%	20%	10%	100%
Q1						
Q2						

Code of Conduct:

I _192565037 / Muthupandi C__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Topic 1: Decision Support Systems (DSS)

Focus Areas:

- Definition and need of DSS
 - Components (Data, Model, UI)
 - Types of DSS
 - Real-world applications (banking, healthcare)
 - Advantages & limitations
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Topic 2: Operational Systems vs DSS

Focus Areas:

1. OLTP vs OLAP concepts
 2. Key differences (data, users, purpose)
 3. Performance comparison
 4. Use-case scenarios
 5. Integration between systems
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Topic 3: Architecture of Data Warehouse

Focus Areas:

1. Data warehouse definition
 2. Three-tier architecture
 3. ETL process
 4. Data marts & metadata
 5. Benefits in decision-making
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Topic 4: OLAP and Data Cubes

Focus Areas:

1. OLAP concept
 2. Data cube structure
 3. OLAP operations (slice, dice, roll-up, drill-down)
 4. Applications in analytics
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Topic 5: Dimensional Data Modeling

Focus Areas:

1. Fact and dimension tables
 2. Star schema vs snowflake schema
 3. Schema design process
 4. Real-world example
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Topic 6: Query Processing & BI Tools

Focus Areas:

1. Query processing stages
 2. Query optimization
 3. BI tools overview
 4. Reporting and visualization
 5. Examples of open-source BI tools
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Topic 7: OLAP Application using KNIME

Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation
-

PART B: PRESENTATION RULES & GUIDELINES

◇ 1. Group Formation

1. Each group: 3–5 students
 2. Each member must participate
 3. Assign roles:
 1. Presenter
 2. Designer (PPT)
 3. Researcher
 4. Demo/Case handler
-

◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
 1. Definition and importance
 2. Core Concepts
 1. Explanation with diagrams
 3. Process/Architecture
 1. Step-by-step explanation
 4. Real-World Example
 1. Case study or application
 5. Tool/Demo (if applicable)
 1. KNIME or BI tool
 6. Conclusion
 1. Summary + future scope
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◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
 1. Diagrams
 2. Flowcharts

3. 3. Tables (for comparison)
 3. Avoid:
 1. Too much text
 2. Copy-paste content
-

◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
 1. Eye contact
 2. Clear voice
 3. Logical flow



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1603 DATA WAREHOUSING AND DATA MINING

The Time Machine of Data

Guide By;
Dr.D.Beulah David

Present By;
Muthupandi C



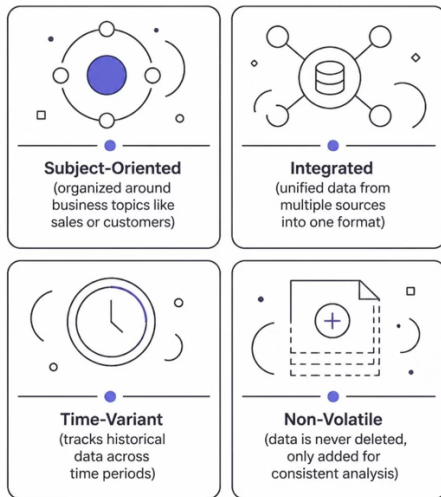
UNIT 1 · INTRODUCTION TO DATA WAREHOUSING

Transforming Raw Data into Smart Business Decisions

- Every day, 2.5 quintillion bytes of data are generated worldwide
- Only 0.5% of all data is ever analyzed — the rest sits unused
- Data warehouses unlock patterns, trends, and predictions hidden in that raw data
- This presentation covers the foundations of data warehousing and how it powers modern business intelligence
- Learn how organizations like Amazon and Netflix turn data into competitive advantage
- **Fun Fact:** A single day's worth of global data could fill every DVD ever made — twice!

What Is a Data Warehouse?

A **data warehouse** is a centralized repository that stores structured data from multiple sources — optimized for analysis, not daily transactions.



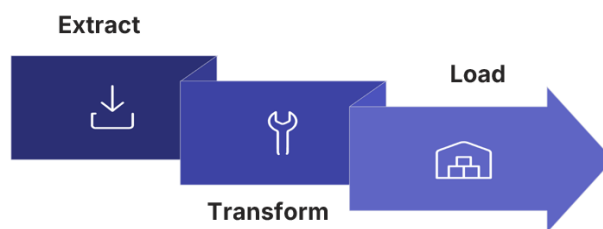
Real-World Example

- **Netflix** stores billions of viewing events to power its recommendation engine
- **Amazon** uses a data warehouse to analyze purchase history and optimize inventory
- Both rely on historical, integrated data — not live transaction logs

🤖 **Think About It:** Why can't a regular database handle analytical queries at scale?

The ETL Process

Extract, Transform, Load — the three-step pipeline that moves raw data into a clean, analysis-ready warehouse.



Why ETL Matters

ETL ensures that only accurate, consistent, and structured data enters the warehouse — making downstream analysis reliable and meaningful.

- Eliminates duplicates and inconsistencies across sources
- Standardizes formats (dates, currencies, units)
- Enables historical tracking and trend analysis
- Reduces query load on source systems
- Supports compliance and data governance

OLTP vs. OLAP

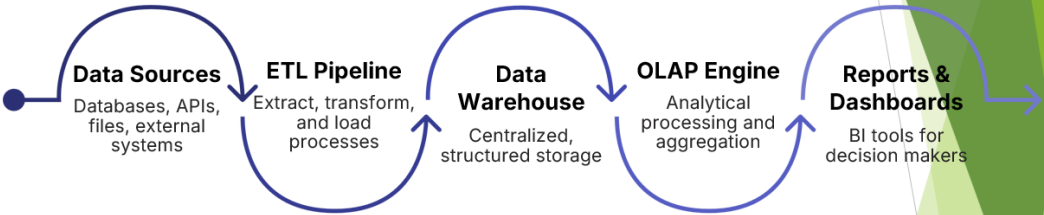
Two fundamentally different systems — one runs your business, the other *understands* it.

Feature	OLTP	OLAP
Purpose	Real-time transaction processing	Complex analytical queries
Data	Current, operational data	Historical, aggregated data
Users	Cashiers, clerks, apps	Analysts, managers, BI tools
Speed	Fast inserts & updates	Fast reads & aggregations
Example	ATM withdrawal, online checkout	Sales trend report, forecasting

  **Quick Quiz:** Which system would you use to *predict next quarter's sales* — OLTP or OLAP?

Data Warehouse Architecture

A layered architecture ensures data flows cleanly from source to insight — with each layer adding structure, quality, and meaning.



Each layer in this pipeline plays a critical role — from raw data ingestion to polished business intelligence.



Banking

Fraud detection, risk modeling, customer segmentation



Amazon & Flipkart

Recommendations, inventory, pricing optimization



Healthcare

Patient outcomes, resource planning, clinical research



Netflix

Content recommendations, viewer behavior, churn prediction

Key Takeaways

Data Warehouse

Centralized, historical, subject-oriented repository for analysis

ETL Pipeline

Extract → Transform → Load ensures clean, consistent data

OLTP vs. OLAP

Transactions vs. analytics — two systems, two purposes

Architecture

Sources → ETL → Warehouse → OLAP → Dashboards

"Data is the fuel, Data Warehousing is the engine, and Business Intelligence is the destination."

Final Quiz: (1) What does the "T" in ETL stand for? (2) Is Netflix's recommendation engine powered by OLTP or OLAP?

Thank You! — Questions welcome. 🎉



Thank You!